Style or Signature? Artist-Disjoint Evaluation of Style Classification in Frozen Vision Embeddings

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Abstract. Frozen image embeddings from models such as CLIP are increasingly used to classify paintings by art-historical style, with high reported accuracy. We ask whether this accuracy reflects an understanding of style or the recognition of individual artists. Standard evaluation uses random train and test splits in which works by the same artist appear on both sides, so a classifier can succeed by recognising the painter rather than the movement. We re-evaluate style classification under an artist-disjoint protocol, holding out every artist in turn so that no work is ever classified using other works by its own painter. On a balanced dataset of 320 paintings across four twentieth-century movements, 5-NN style accuracy falls from 0.87 to 0.77 under this protocol, and the drop is sharply uneven. Impressionism and Cubism barely move, while Surrealism falls twenty points. The pattern holds across four image encoders, including a vision-only self-supervised model, which rules out a language-driven explanation. Where an encoder captures genuine shared form, individual artists are barely recognisable yet style is robust; Surrealism shows the opposite. We argue that artist-disjoint evaluation is necessary to measure stylistic understanding in frozen embeddings.

Keywords: computational art history · CLIP · evaluation · style classification

1 Introduction

Frozen image embeddings from large pretrained vision models, CLIP [11] foremost among them, are increasingly used off the shelf to analyse visual art, including organising collections, retrieving similar works, and classifying paintings by art-historical style [1, 4]. A common and encouraging finding is that style is highly recoverable from these embeddings, since a simple nearest-neighbour or linear classifier on frozen features can assign movement labels with high accuracy and no training on art, as we ourselves find (Section 5.1). It is tempting to read such accuracy as evidence that these encoders have learned something about artistic style as a by-product of web-scale pretraining.

We argue that this reading is premature, because the standard way of measuring it cannot distinguish style from a confound these embeddings encode strongly, the identity of the individual artist. Style-classification studies typically evaluate on a random split of works, in which paintings by a given artist may fall on both the reference and the query side [6]. A classifier can then label a new Dalí as Surrealist either because it recognises Surrealism or because it recognises Dalí, and the evaluation cannot tell which. This would be hypothetical if artist identity were weakly represented, but it is not. Frozen CLIP features support artist identification at high accuracy, a signal strong enough to have been described as a neural artist signature [8]. Whenever same-artist works are available as neighbours, a style score may therefore be measuring artist recognition in part, and a single aggregate accuracy gives no way to see how much, or for which movements.

The remedy is to control for artist identity directly, evaluating on artist-disjoint splits in which no work is classified using other works by its own painter. This is standard hygiene in other domains, where models are tested on held-out patients or speakers to prevent identity leakage from masquerading as task performance, and a recent survey of style classification recommends artist- or source-disjoint splits and examining *why* a style label is assigned rather than only whether a model is accurate [10]. To our knowledge this control has not been applied to style classification in frozen vision embeddings. We apply it and find that it changes the conclusions substantially.

On a balanced dataset of 320 paintings spanning four twentieth-century movements, replacing the standard evaluation with an artist-disjoint one lowers 5-NN style accuracy from 0.87 to 0.77. The aggregate drop is modest but averages over a sharply uneven effect. Impressionism and Cubism barely move, while Surrealism falls twenty points, near the level at which the model misclassifies half its works (Section 5.2). An artist-recognisability control explains the unevenness (Section 5.3). The robust movements are those whose individual artists the embedding can least tell apart, the mark of a shared visual form that has subsumed individual authorship, whereas Surrealism combines higher artist-recognisability with the lowest robustness, its collapse concentrated in specific painters whose style scores were borrowing from their recognisability as individuals. The pattern is not specific to CLIP. It holds across four encoders differing in scale, architecture, and training objective, including a vision-only self-supervised model whose agreement rules out any language-derived explanation (Section 5.4).

Contributions.

1. We introduce an **artist-disjoint (leave-one-artist-out) evaluation protocol** for painting-style classification in frozen vision embeddings, holding whole artists out of the neighbour pool, and show that it exposes fragility which standard random-split evaluation conceals.
2. We quantify the gap between standard and artist-disjoint accuracy, show that it is **sharply movement-dependent** rather than uniform, and identify its mechanism with an **artist-recognisability control** that separates genuine shared form from artist memorisation, and further separates memorised artists from those the embedding simply represents poorly.
3. We show the effect is **encoder-general** across four image encoders, including a vision-only model whose agreement rules out a text-contrastive explanation.

2 Related work

Frozen vision embeddings for art. General-purpose vision encoders, CLIP [11] in particular, are widely used off the shelf to analyse paintings, on the premise that web-scale pretraining yields features that transfer to art without adaptation. Recent work probes this premise. Asperti et al. [1] report that CLIP captures semantic content more readily than stylistic nuance, and work on predicting Wölfflin's principles finds that off-the-shelf CLIP struggles with fine-grained style and requires fine-tuning [4]. We ask a different question, not how well frozen embeddings classify style in absolute terms, but whether that accuracy measures style at all once the confounding contribution of artist identity is removed.

Style classification on WikiArt. Classifying paintings by movement is an established task, typically posed on WikiArt and evaluated by training a classifier and testing on a held-out split of the same corpus [6]. A recent survey observes that many studies train and evaluate within the same dataset, making it hard to tell whether models learn transferable stylistic features or dataset-specific artefacts, and calls for duplicate removal, artist- or source-disjoint splits, and transparent class distributions [10]. Our protocol implements that recommendation.

Artist identity as a confound. That artist identity is strongly encoded in these representations is well established. Moayeri et al. [8] show that a classifier on frozen features matches works to their artist for 89.3% of 372 WikiArt artists, describing these recurring characteristics as neural *signatures*. This is the signal a random-split style evaluation leaves uncontrolled, since if a movement's works are recognisable by their individual painters, a classifier can reach the movement label by way of the artist. We treat it as the confound to be removed and measure how much of a movement's separability survives its removal.

Group-disjoint evaluation. Holding out a grouping variable to prevent identity leakage is established practice in adjacent fields. In medical imaging, the standard practice is to split at the patient level, so that a model cannot appear to detect a condition when it is actually recognising a patient it has already seen [12]. In speech recognition, the common strategy is to hold out whole speakers, with the observation that which speaker is held out can drive substantial variability [7], a parallel to the per-artist variation we report. Our contribution is not this principle but its application to painting-style classification in frozen vision embeddings, where to our knowledge it has not been used.

3 Data

We use a *style dataset* of 320 paintings spanning four twentieth-century movements, built to study how cleanly frozen image embeddings separate art-historical styles and whether that separation survives when whole artists are held out.

3.1 Style dataset

Source and shape. The style dataset is drawn from the `Artificio/WikiArt` release on Hugging Face, a static snapshot of WikiArt. We sample from a frozen, versioned release rather than scraping the live site, so that the precise set is reproducible and cannot drift between experiments, and because this release assigns a single style label per work, avoiding the multi-tagging inflation of the live site where one work (Picasso's especially) may carry several movement tags. The subset is balanced by construction, with four movements, eight artists per movement, and ten works per artist, giving 320 images. Because our central experiment holds out whole artists (Section 4), the number of *artists* per class governs the strength of the generalisation estimate, and equal per-artist counts remove a potential artist-frequency imbalance.

Artist selection. Each movement is represented by the eight artists most strongly associated with it in standard art-historical references [14], filtered to those on the movement's roster in this release with at least ten qualifying works. We cite the selection *criterion* rather than each artist, since the uncontested core of each movement (such as Monet, Picasso, and Dalí) follows from any standard reference. We tested an automatic ranking of roster members by tagged-work count and rejected it, since that count tracks digitisation volume rather than centrality, placing Sam Francis above Pollock and admitting Matisse to Impressionism. Work count is therefore used only as an inclusion filter. Four decisions where canon is contested or a source constraint forced a non-obvious choice are documented under *Margin decisions*.

The eight artists per movement are Pollock, Rothko, Krasner, Kline, Newman, Still, Hofmann and Gorky (Abstract Expressionism); Picasso, Gris, Léger, Braque, Metzinger, Gleizes, Marcoussis and Villon (Cubism); Monet, Renoir, Degas, Sisley, Pissarro, Cassatt, Morisot and Caillebotte (Impressionism); and Dalí, Magritte, Ernst, Miró, Delvaux, Masson, Carrington and Tanguy (Surrealism).

Work selection. For each artist we drew ten works at random from those tagged with any label in the movement's label set (listed below), above a resolution floor and with a fixed per-artist seed. In practice this release stores uniformly sized thumbnails, so the floor had no effect and the step reduced to a uniform random draw over each artist's eligible works. Resolution thus acts as a quality gate, not a selection criterion, so the subset is not biased toward an artist's most famous or most digitised works.

Label sets. Two of the four movements are umbrellas that this release shards into sub-styles, so we pool labels for those two and use single labels for the other two. Cubism pools Cubism, Analytical Cubism and Synthetic Cubism, which are phases of one movement. Abstract Expressionism pools Abstract Expressionism, Action Painting and Color Field Painting, the movement's two recognised wings [13]. This pooling is *artist-gated*. Because selection is by named artist, pooling Color Field into Abstract Expressionism recovers first-generation figures such as Newman and Still without admitting the second-generation post-painterly painters (Louis, Noland, Olitski) who carry the same tag, and the same gating makes the Cubism merge safe. This is the dataset's one structural asymmetry. Artist-gating does make class membership partly artist-defined in a study aiming to separate style from artist identity, a tension we revisit in Section 7.

Preprocessing. Images are stored as 256×256 thumbnails produced by anisotropic resizing, a non-uniform squeeze that distorts aspect ratio but crops no content, so framing, composition and the figure-to-ground boundaries that matter for formal style are retained. The distortion is applied uniformly, though we cannot rule out differential effects on geometry-dependent movements, and every encoder in our comparison receives the identically distorted input. It does mean the images are not quite what these encoders saw at pretraining time, which we revisit in Section 7.

Near-duplicate check. WikiArt occasionally stores a work twice, or a detail crop alongside a full reproduction, so we screened for duplicates, embedding all 320 images with CLIP ViT-B/32 and computing cosine similarity for every pair. No pair reached the 0.95 flagging threshold. The closest pair (two works by Joan Miró, cosine 0.949) and the next-closest band (works by Gorky, Rothko and Pollock at 0.93 to 0.94) were inspected and confirmed distinct. Notably, the highest similarities are between distinct works by the same artist rather than repeated reproductions. The frozen dataset and its manifest are versioned by content hash, so no image can change silently between experiments.

Data availability. The 320 works are pinned by their index in the `Artificio/WikiArt` release, so the exact set is reproducible regardless of the selection procedure. Reconstruction depends on this release rather than the live `wikiart.org` site, since an attempted re-fetch from the live site by title-matching succeeded for only 235/320 (73%), mostly because of title-encoding mismatches and distinct works sharing a title. The pinned indices avoid this ambiguity.

3.2 Source limitations

Several properties of the source bear on interpretation and shape the data itself. WikiArt's labels are crowd-sourced rather than expert-assigned, and the full corpus is imbalanced across artists and movements. "Most strongly associated with the movement" is not a neutral criterion, since it reflects which artists were historically collected, exhibited and digitised, and WikiArt inherits that skew. Our own choices participate in it. Excluding Kahlo (below), for instance, removes one of the few non-Western-centred figures in the pool on grounds of contested membership. The selection therefore captures the conventional canon of each movement rather than an objective or complete membership.

3.3 Margin decisions

Four selection decisions warrant note. *(i) Kahlo excluded from Surrealism*, because although Breton championed her, she rejected the label [2] and is widely read as closer to folk or magic realism, so we selected Tanguy, Delvaux, Masson and Carrington to keep the class to undisputed members. *(ii) Three canonical Abstract Expressionists replaced*, since de Kooning, Motherwell and Gottlieb are absent from this release, so we substituted Newman, Still and Hofmann, all first-generation New York School figures. *(iii) Abstract Expressionism pools three sub-styles*, artist-gated so there are no second-generation post-painterly painters. *(iv) Cubism pools three sub-styles*, with Robert Delaunay dropped (his work is tagged Orphism, a distinct movement) with Louis Marcoussis replacing him.

4 Method

Embeddings. Every painting is encoded once by a frozen, pretrained image encoder and reused across all analyses, with no fine-tuning at any point. Each image is opened in RGB, passed through the encoder's standard preprocessing, and embedded with gradients disabled. The feature vector is L_2-normalised, so cosine similarity reduces to an inner product. Unless stated otherwise, results use CLIP ViT-B/32 [11] as the reference encoder, and Section 5.4 repeats the full pipeline with CLIP ViT-L/14, the self-supervised DINOv2 ViT-B/14 [9] (CLS token), and a supervised ImageNet ResNet-50 [5] (global-average-pooled). All nearest-neighbour computations use cosine distance.

Full-pool style classification. As a descriptive measure of how the dataset clusters by style, we run 5-NN classification over all 320 works, assigning each work the majority style among its five nearest neighbours excluding itself, with ties broken by the closest neighbour then alphabetically. We use $k = 5$ throughout, and accuracy is stable across $k \in \{3, 5, 7, 9\}$. As a classifier-free view we report mean silhouette scores using the style labels as cluster assignments.

Artist-disjoint style classification. The full-pool measure, like standard random-split evaluation, permits other works by a query's own artist to serve as neighbours, so it cannot separate movement-level structure from individual-artist recognition. We control for this with a leave-one-artist-out protocol. Each of the 32 artists is held out in turn, that artist's ten works become queries, and the remaining 310 works form the reference pool, so no work is ever classified using another by the same painter. The protocol is exhaustive and deterministic. Because each movement contributes eight held-out artists, we also report the standard deviation across those eight folds.

Artist-recognisability control. To test whether style success is partly artist recognition, we re-run the same procedure but predict the *artist* rather than the style, with same-artist works permitted as neighbours, measuring how recognisable each artist is individually. We report overall accuracy against the 1/32 chance rate, the mean per movement, and the per-artist values, set against the artist-disjoint style accuracies of the same artists.

5 Results

5.1 Full-pool separability

A 5-NN classifier over all 320 works recovers the four style labels with 0.869 accuracy overall, well above the 0.25 chance and majority-class rates. Per-style accuracy ranges from 0.975 for Abstract Expressionism down to 0.713 for Surrealism, with Cubism at 0.850 and Impressionism at 0.938. Silhouette scores are more cautious about cluster separation. The overall mean silhouette is only weakly positive (0.066), and Surrealism is the sole movement with a negative mean (-0.039), meaning many Surrealist works lie closer to other movements than to their own. Two features of the errors anticipate later results. Abstract Expressionism behaves as a sink, receiving more misclassifications than it produces, consistent with its non-representational character, and Surrealism is the movement whose separability we will find to depend most heavily on individual artists. These numbers are the baseline that the artist-disjoint protocol of Section 5.2 re-examines.

5.2 Artist-disjoint evaluation

The full-pool measure cannot separate two abilities, recognising a *movement* from its shared visual form and recognising an individual *artist* whose other works happen to share the query's label. Because these encoders carry a strong artist signal (Section 5.3), a high full-pool score is consistent with either. To force the distinction we apply the artist-disjoint protocol of Section 4, holding each of the 32 artists out in turn so that every painting is classified exactly once, when its own painter contributes nothing to the reference pool.

Aggregate accuracy falls, and the fall is uneven. Overall 5-NN style accuracy drops from 0.869 to 0.766 under the artist-disjoint protocol (Table 1). That ten-point gap is the share of the full-pool score that depended on the model seeing other works by the query's own artist. The per-style breakdown shows the reliance is far from uniform. Impressionism and Cubism barely move, falling 3.8 and 5.0 points and remaining highly separable without same-artist neighbours. Surrealism falls 20.0 points, from 0.713 to 0.513, the only movement for which the model misclassifies close to half of all held-out works. Abstract Expressionism sits between, dropping 12.5 points while staying high at 0.850. The spread, rather than the average, is what matters here. Some movements possess artist-invariant visual form that these embeddings capture, and others do not.

Table 1: Style classification accuracy under the full-pool protocol (all works eligible as neighbours) and the artist-disjoint protocol (each artist held out in turn). The last column is the standard deviation across the eight held-out artists of each movement.

Style	Full-pool	Artist-disjoint	Drop	std (across artists)
Abstract Expressionism	0.975	0.850	0.125	0.250
Cubism	0.850	0.800	0.050	0.187
Impressionism	0.938	0.900	0.038	0.132
Surrealism	0.713	0.513	0.200	0.226
Overall	0.869	0.766	0.103	—

Variance across held-out artists is itself diagnostic. With only eight artists per movement, a single held-out painter can move a per-style figure substantially, so the standard deviation across the eight folds (Table 1, last column) carries information a mean hides. A movement whose accuracy is stable regardless of which artist is removed coheres through shared form, while one whose accuracy swings with the held-out artist is held together by the individual painters present. Impressionism has both the smallest drop and the smallest variance (0.132), the signature of a genuinely shared style. Surrealism combines the largest drop with high variance (0.226), and the per-artist view (Figure 1) shows why. Its collapse is concentrated in particular artists rather than spread evenly. Two automatist painters, Miró and Masson, fall to 0.1 and 0.2 when their own works are removed, while the other six Surrealists stay between 0.5 and 0.8. Surrealism's fragility is thus driven by a minority of its artists, which Section 5.3 takes up.

Where held-out works misroute. The errors are not random but concentrate in two destinations. Abstract Expressionism acts as a sink, receiving far more misroutes than it emits, consistent with its non-representational character drawing in any sufficiently abstract work. Surrealism's 39 errors split almost evenly between Abstract Expressionism and Cubism, the abstract automatist works drifting toward the former and the more geometric compositions toward the latter. Surrealism does not dissolve into noise when its artists are held out. It is pulled apart along two interpretable visual axes, neither of which is the Surrealist label.

5.3 The mechanism: artist recognisability

Section 5.2 showed that holding artists out lowers style accuracy unevenly and that Surrealism's fall is concentrated in particular painters. To ask what those painters have in common, we run the same nearest-neighbour procedure but predict the *artist* rather than the style, with same-artist works permitted as neighbours, which measures how recognisable each artist is as an individual. If the embedding encodes who painted a work, a classifier can reach the correct movement by way of the artist and will fail only when that artist is absent.

The artist signal is strong. Across the 320 works, 5-NN artist classification reaches 0.466 against a chance rate of $1/32 = 0.031$, identifying the individual

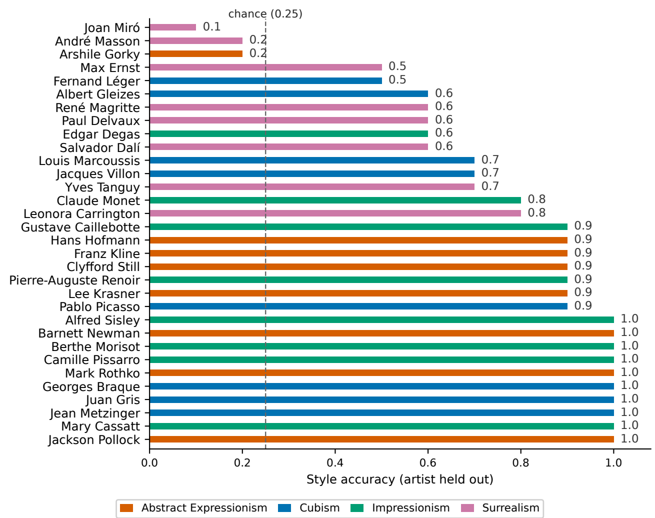

Fig. 1: Per-artist style accuracy under the artist-disjoint protocol, all 32 artists sorted ascending and coloured by movement. The dashed line marks the four-class chance rate (0.25). The lowest scores are the automatist Surrealists Miró (0.1) and Masson (0.2), with Gorky (0.2, Abstract Expressionism) the only non-Surrealist among them.

painter roughly fifteen times more often than chance. Artist identity is therefore heavily represented and available to inflate style scores wherever same-artist neighbours are present. Whether it actually drives the style results is answered by the per-artist structure.

Recognisability and style robustness are not simply related. One might expect a clean inverse relationship, the more recognisable an artist the more their style score should collapse when held out. Across the 32 artists this relationship is essentially absent (Pearson $r = 0.06$), because all combinations occur. Some artists are recognisable and classify well by style, some are recognisable and collapse, some are unrecognisable yet classify well, and some are weak on both. A single correlation averages these away. The informative structure is a set of distinct artist types, visible when recognisability is set against artist-disjoint style accuracy.

Three types of artist. The artists who drive Surrealism's collapse do not form one category. *Memorised* artists are highly recognisable yet cannot be placed by style once held out. Gorky is recognised as himself at 0.90 but classified by style at 0.20, and Miró is recognised at 0.50 yet placed by style at 0.10. For these painters the embedding encodes the individual hand, and the earlier style success was borrowing from it. *Poorly represented* artists are weak on both measures. Masson is recognised at only 0.30 and classified by style at 0.20, so his failure is not memorisation unmasked but a basic difficulty the embedding has with his

Table 2: Mean artist-recognisability per movement (5-NN artist classification, same-artist neighbours permitted) against artist-disjoint style accuracy. Cubism and Impressionism combine low recognisability with high robustness; Surrealism combines higher recognisability with the lowest robustness.

Movement	Artist-recognisability	Artist-disjoint style acc.
Cubism	0.238	0.800
Impressionism	0.400	0.900
Abstract Expressionism	0.713	0.850
Surrealism	0.513	0.513

work. The distinction matters, since Miró and Masson, the two Surrealists who fall furthest in Section 5.2, fail for different reasons. The third type is best seen at the level of movements.

Shared-form movements subsume the individual. Aggregating recognisability by movement (Table 2) recovers the contrast that holds across encoders (Section 5.4). Cubism has the lowest mean artist-recognisability, 0.24, since the encoder struggles to tell Picasso, Gris, Braque and their peers apart, unsurprising given that in the Analytical phase the works of Picasso and Braque became famously hard to tell apart as authorship gave way to a shared concern with the structure of form [3]. Yet Cubism is among the most robust movements under hold-out, and Impressionism behaves similarly. Surrealism is the mirror image, its artists more individually recognisable (0.51) yet its style the least robust. Where the encoder has learned a shared visual form, individual authorship is partly dissolved into it and the movement classifies robustly because it depends on no one artist. Where no shared form is available, the movement is a loose collection of individuals, and classification is only as durable as their presence in the pool.

A note on Abstract Expressionism. Abstract Expressionism fits neither pole cleanly. Its artists are the most individually recognisable of any movement (0.71), yet the movement stays fairly robust under hold-out (0.85). Distinctive individual styles and a coherent movement-level signal can coexist, since Pollock, Rothko and Newman are each immediately recognisable and each classify perfectly under hold-out. The exception is Gorky, whose collapse is masked at the movement level by his robust colleagues. The memorisation failure can thus occur inside an otherwise-cohesive movement without dragging the class down.

5.4 Generalisation across encoders

To test whether the pattern reflects an idiosyncrasy of CLIP ViT-B/32 or of image–text training, we repeat the full pipeline across four encoders differing in scale, architecture, and objective, the two CLIP models ViT-B/32 and ViT-L/14, the self-supervised vision-only DINOv2 ViT-B/14, and a supervised ImageNet

Table 3: Per-style accuracy under the artist-disjoint protocol across four encoders. Absolute accuracies are not comparable across encoders; the consistent pattern is that Surrealism is lowest in every row (**bold**), while Impressionism and Cubism remain robust.

Encoder	Abs. Expr.	Cubism	Impressionism	Surrealism
CLIP ViT-B/32	0.850	0.800	0.900	**0.513**
CLIP ViT-L/14	0.950	0.838	0.925	**0.638**
DINOv2 ViT-B/14	0.913	0.763	0.875	**0.375**
ResNet-50 (ImageNet)	0.538	0.763	0.813	**0.500**

ResNet-50. Comparisons are made at the level of pattern rather than absolute accuracy, since the encoders differ in embedding dimension.

The fragility is encoder-general. Surrealism is the most fragile movement under the artist-disjoint protocol for every encoder (Table 3), with the lowest accuracy of the four movements without exception, at 0.513, 0.638, 0.375, and 0.500. The effect survives a change of scale within the CLIP family, where the larger ViT-L/14 lifts every accuracy but leaves Surrealism lowest, a change of architecture from transformer to convolutional network, and the move to a supervised encoder that is neither contrastive nor self-supervised. This is best described as a property of the movements, not of CLIP. Surrealism lacks the artist-invariant visual regularity the other movements possess, and no encoder here recovers it.

The effect is visual, not linguistic. DINOv2 is the most informative case. Trained by self-supervision on images alone, with no text encoder, it cannot exploit any correspondence between a painting and a verbal description of its style. If the Surrealism collapse arose from CLIP's language side, DINOv2 should not reproduce it. It reproduces it more sharply than any CLIP model, with Surrealism at 0.375. The fragility is therefore in the visual structure of the movements, not in language-derived associations.

Shared form subsumes individual identity. Pairing the artist-disjoint accuracies with the artist-recognisability scores of Section 5.3 exposes a consistent movement-level relationship. The two robust movements, Cubism and Impressionism, are also those whose artists the encoders find hardest to tell apart, lowest in recognisability yet among the highest in robustness. Surrealism shows the opposite. This is the clearest evidence that style separability and artist recognition are distinct. Where an encoder captures a shared visual form, as in Cubism, it classifies the movement robustly because authorship has been subsumed into that form. Where none is available, as in Surrealism, the encoder falls back on individual-artist signal and fails when that signal is removed. One minor departure does not disturb this, that under DINOv2 Impressionism rather than Cubism has the marginally lowest recognisability, a reordering within the shared-form pair.

6 Discussion

Style separability is not one thing. Our results argue against reading a single style-classification accuracy as a measure of how well an embedding "understands" style. The aggregate figure of 0.87 conceals a range from 0.51 to 0.90 once artists are held out, and that range is the substantive finding. The four movements are separable for different reasons. Impressionism and Cubism are held together by visual form shared across their members and survive the removal of any single artist, whereas Surrealism is held together loosely, with much of its apparent separability resting on the embedding recognising particular painters. "Does the embedding capture style?" is therefore the wrong question, since it presupposes that style is a uniform property. The better question, which our protocol operationalises, is for which movements, and through what, a style label is recoverable once individual authorship is controlled for.

Shared form versus artist lookup. The artist-recognisability control names the two ends of this spectrum. At one end, a movement classifies robustly *because* its artists are hard to tell apart. Cubism has the lowest artist-recognisability of any movement yet among the highest robustness under hold-out, the signature of a genuine shared form that has subsumed individual handwriting. This is the cleanest rebuttal to the suspicion that a style classifier is merely an artist classifier in disguise, since for Cubism the encoder cannot reliably identify who made a work yet still places it as Cubist. At the other end, a movement can appear separable because the embedding has memorised its members, which Surrealism approaches. The per-artist view sharpens this, since Gorky and Miró are recognised as individuals far more reliably than they are placed by style, so for them the style signal was substantially a borrowed artist signal. A third pattern must be kept separate, the poorly represented artist, exemplified by Masson, who is weak on both measures and whose failure reflects a basic difficulty the embedding has with his work rather than memorisation unmasked. The two Surrealists who fall furthest therefore fail for different reasons.

A property of the movements, not of CLIP. That the same ordering appears across four encoders differing in scale, architecture, and objective indicates a property of the movements as rendered in pixels rather than an artefact of one model. The DINOv2 result is decisive on the question of language, since a vision-only encoder with no access to text reproduces the Surrealism collapse as sharply as any CLIP model, ruling out explanations that route through image–text alignment or through movement names co-occurring with images on the web. Whatever makes Surrealism hard is visible to encoders that never read the word "Surrealism". This fits art-historical intuition. Impressionism and Cubism are defined substantially by how their surfaces look, broken colour and dissolved light in one case and the fragmentation of form into geometric facets in the other, whereas Surrealism is unified far more by a shared idea, the staging of dream logic and incongruous juxtaposition, than by any shared visual handling. A representation built from visual statistics captures the former kind of unity and largely misses the latter, and our measurements are what that mismatch looks like quantitatively.

Implications for evaluation. The practical consequence is that random-split evaluation can substantially overstate how well a model captures style, by up to twenty points for one movement here and invisibly in the aggregate. Because frozen embeddings are now routinely used off the shelf to organise and retrieve artworks, an evaluation that allows same-artist works on both sides of the split risks certifying as "style understanding" what is partly artist recognition, unevenly across movements in a way a single figure hides. Artist-disjoint evaluation is inexpensive, needs no retraining, and turns a brittle aggregate into an interpretable per-movement and per-artist picture, so we would suggest it as a default rather than a refinement when the question is genuinely about style. The principle is not novel, since group-disjoint evaluation is standard in domains such as medical imaging and speaker-independent speech recognition. Our contribution is to bring it to painting-style classification, where a recent survey observes it has been largely absent [10], and to show how much it changes the conclusions.

Scope and an encoder-specific aside. Two cautions bound these claims. First, absolute accuracies are not comparable across encoders, so we lean only on the consistent ordering. One departure is that the ImageNet-trained ResNet-50 classifies Abstract Expressionism far less robustly than the transformer encoders, plausibly because a convolutional network trained for object recognition keys on local texture, and gestural abstraction offers texture without the objects such a model expects. Second, our dataset is small and its movements are not matched for subject matter, so some of what we attribute to style remains entangled with content and composition, which we take up in Section 7. Neither disturbs the central result, which is comparative and internal to the dataset. Whatever the residual confounds, they apply across movements and cannot explain why holding out artists costs Surrealism twenty points and Impressionism four.

7 Limitations

Scale and balance. The style dataset is small: 320 works, eight artists per movement, ten works per artist. A single held-out painter moves a per-style figure by up to one eighth, which is why we report the spread across held-out artists alongside the mean and treat individual artist effects, rather than tight confidence intervals, as the appropriate unit of evidence. The balance aids interpretation, but the absolute accuracies should be read as indicative for this curated set rather than as estimates of population performance.

Content is not controlled. The movements are not matched for subject matter or composition, so style remains entangled with content: Impressionist landscapes and Cubist still lifes differ in what they depict as well as how. Some of what we attribute to style separability may therefore reflect subject and composition.

This does not undermine the central, comparative result, since any such confound applies across movements and cannot by itself explain why holding out artists costs Surrealism twenty points and Impressionism four, but it does mean the per-movement accuracies should not be read as measuring style in isolation.

Labels, canon, and preprocessing. The dataset inherits properties of its source (Section 3). Labels are crowd-sourced rather than expert-assigned; the selection captures the conventional, Western- and male-skewed canon of each movement rather than an objective membership, a skew our own choices participate in; and the style-dataset images are anisotropically resized to 256×256, distorting aspect ratio, though without cropping content and uniformly across all movements and encoders, so the distortion cannot differentially explain any per-movement result.

Scope of the probe. We study frozen embeddings used as fixed feature extractors with nearest-neighbour classification, a deliberately simple and transparent setting. We do not fine-tune, probe intermediate layers, or use the encoders' text towers, and our claims are correspondingly scoped: they concern what off-the-shelf image embeddings make linearly accessible by nearest neighbour, not the limits of what these models could represent after adaptation. Fine-tuning on art-specific labels is known to improve stylistic sensitivity, and how artist dependence behaves under such adaptation is left to future work.

8 Conclusion

We re-examined a familiar result, that frozen vision embeddings classify paintings by style with high accuracy, under an evaluation that controls for the artist. Holding out whole artists lowers 5-NN style accuracy from 0.87 to 0.77, and the modest aggregate change conceals a sharply uneven effect: Impressionism and Cubism are essentially unaffected, while Surrealism falls twenty points. An artist-recognisability control accounts for the unevenness. The robust movements are those whose individual artists the embedding can least tell apart, indicating a genuine shared visual form that has subsumed individual authorship, whereas Surrealism's separability rests substantially on the recognisability of particular painters and collapses when they are removed. The pattern holds across four encoders differing in scale, architecture, and objective, including a vision-only model whose agreement places the effect in visual structure rather than language.

The broader point is methodological. A single style-classification accuracy can present as understanding what is partly artist recognition, and can do so unevenly across movements in a way the aggregate hides. Artist-disjoint evaluation is inexpensive, needs no retraining, and turns that aggregate into an interpretable per-movement and per-artist picture; we would encourage its use as a default wherever the question is genuinely about style rather than about artists. Natural extensions include content-controlled subsets that match subject across movements, probing the text encoders alongside the image ones, and asking whether fine-tuning closes or merely relocates the artist dependence we have measured.

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